

LESSON 5-6 • HOMEWORK PACKET • SHEET 1 OF 2

Drawing Nets of Solids

Maryland Standard 6.G.A.4 • Learning target – I can draw the net of a solid by unfolding its faces onto a flat surface.

REMEMBER

- A **net** is a flat pattern that folds into a solid. Each face appears **once**, with no overlaps.
- Rectangular prism:** 6 rectangles. **Square pyramid:** 1 square + 4 triangles. **Triangular prism:** 2 triangles + 3 rectangles.

WORKED EXAMPLE

Plan a net for a box 5 cm × 3 cm × 2 cm.

- Step 1** Faces come in pairs: 5 × 3, 5 × 2, 3 × 2
- Step 2** Draw the 5 × 3 base. Attach a 5 × 2 face to each long edge.
- Step 3** Attach a 3 × 2 face to each short edge. Add the 5 × 3 top to one 5 × 2 face.

Answer 6 rectangles, two of each size

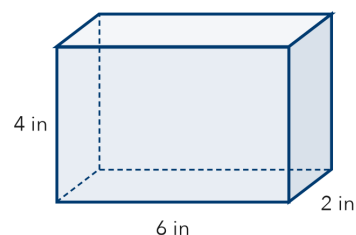
PRACTICE

show your work • label units

1. Draw a net for this box. Label the length and width of every face.

Draw the net here.

Face sizes _____



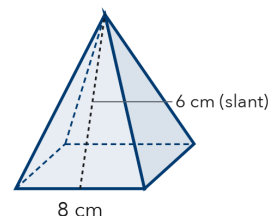
2. How many of each shape are in the net?

- a. cube: squares ____
- b. square pyramid: triangles ____
- c. triangular prism: rectangles ____
- d. pentagonal pyramid: triangles ____

3. The net of this square pyramid has a square and 4 triangles.

- a. Area of the square _____ cm²
- b. Area of one triangle _____ cm²

All 4 triangles _____ cm²



4. **Cereal Box.** A cereal box is 8 in long, 3 in wide, and 12 in tall. List the sizes of the 3 pairs of matching faces in its net.

Face pairs _____

5. **Will It Fold?** Jamal draws 6 squares in one long row and says it is a net for a cube. Is he right? Explain.

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Open an empty cereal box along its edges. Show someone its net.

LESSON 5-6 • HOMEWORK PACKET • SHEET 2 OF 2

Reading Nets and Solids

Maryland Standard 6.G.A.4 • Learning target – I can name the solid a net makes by reading the shapes in the net.

REMEMBER

- **Prism:** 2 matching bases + rectangles. **Pyramid:** 1 base + triangles that meet at a point.
- The **base shape** names the solid (triangular, square, pentagonal ...).

WORKED EXAMPLE

A net has 1 hexagon and 6 triangles.

Step 1 One base with triangles around it means a pyramid.

Step 2 The base is a hexagon.

Answer hexagonal pyramid

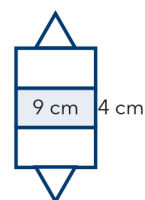
PRACTICE

show your work • label units

1. Look at the net.

a. Solid: _____

b. Number of faces: ____



2. Name the solid each net makes.

a. 1 rectangle + 4 triangles: _____

b. 2 hexagons + 6 rectangles: _____

c. 2 triangles + 3 rectangles: _____

d. 1 pentagon + 5 triangles: _____

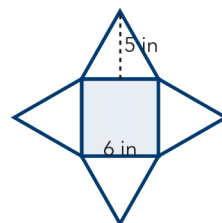
e. 6 squares: _____

f. 2 pentagons + 5 rectangles: _____

3. This net folds into a square pyramid.

a. Area of the base: _____ in²

b. Area of all 4 triangles: _____ in²



4. **Mystery Box.** A net has 2 squares, 4 cm on each side, and 4 rectangles, 4 cm × 9 cm. Name the solid and its dimensions.

Answer _____

TEST PRACTICE

choose the best answer

5. A triangular prism has 5 faces. How many edges does it have?

A 5

B 6

C 9

D 12

6. How many faces does a hexagonal pyramid have?

A 6

B 7

C 8

D 12

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about # _____

AT HOME Find a box at home. Tell someone what shapes its net would have and how many of each.